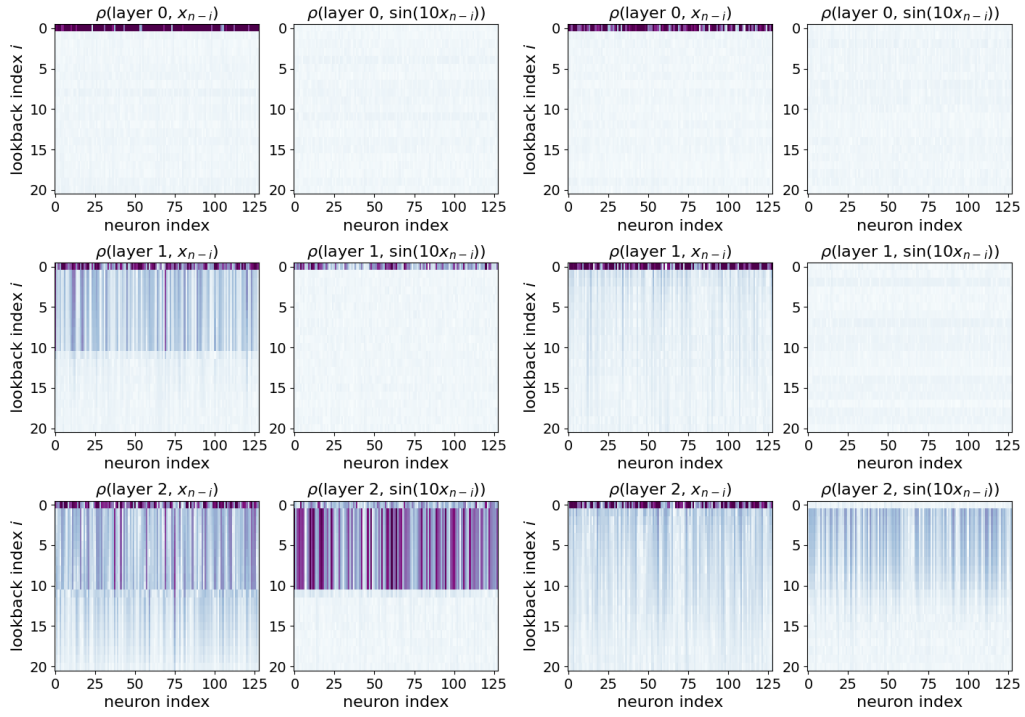


Figure 3: Empirical correlations between each hidden state neuron and x_{n-i} or $\sin(bx_{n-i})$. Models are vanilla (left two columns) and myopic (right two columns) transformers trained on $\mathcal{D}_{0.3}$.

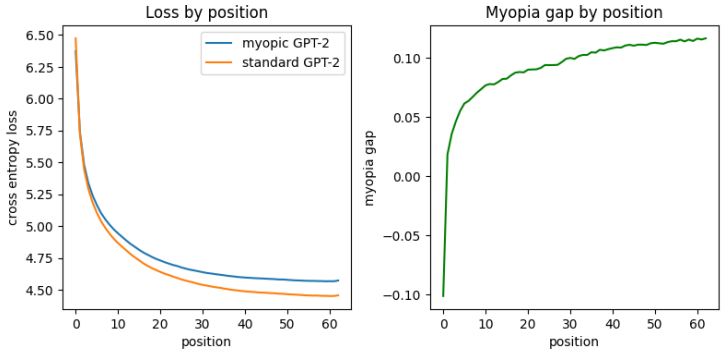
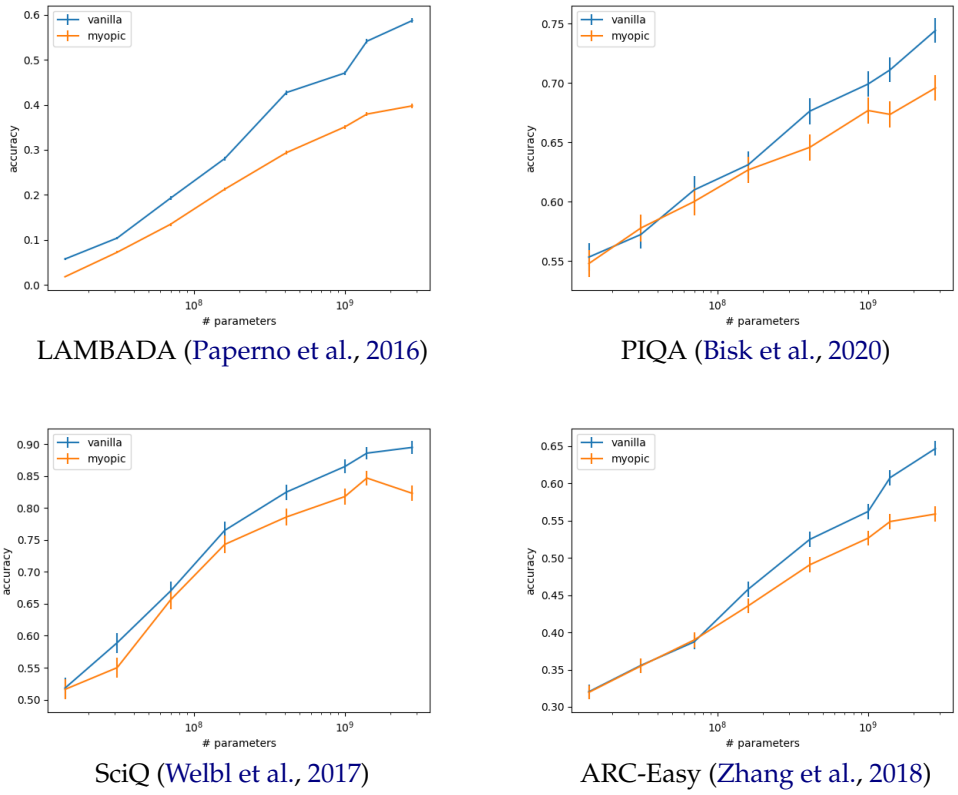


Figure 4: Cross-entropy loss of vanilla and myopic GPT-2 models by token position, and their difference. Evaluated on a sliding window over a 100K-token sample text from the PG-19 dataset (Rae et al., 2019). Aggregate cross-entropy losses on this sample are 4.67 (vanilla) and 4.77 (myopic).

5.2 Myopia gap scaling

One might suppose that the relatively small myopia gap of GPT-2 is due to the relative simplicity of the small architecture we consider, and that larger language models might exhibit a more pronounced myopia gap.

To test this, we train both vanilla and myopic transformers from the Pythia LLM suite (Biderman et al., 2023), ranging in size from 14M to 2.8B parameters, on one epoch of 10M



LAMBADA (Paperno et al., 2016) PIQA (Bisk et al., 2020)

SciQ (Welbl et al., 2017) ARC-Easy (Zhang et al., 2018)

Figure 5: Benchmarks of Pythia models fine-tuned on the Pile dateset using vanilla and myopic descent.

sequences of 64 tokens each subsampled from the Pile dataset (Gao et al., 2020). (We use the same subsampled dataset for every training run.) We report validation cross-entropy loss (Figure 7 in Appendix D) as well as performance on a variety of natural language experiments (Figure 5). Note that, unlike in the GPT-2 experiments (Section 5.1), which start from random initialization, we start all training for Pythia models from the pre-trained checkpoints provided by Biderman et al. (2023)—for the larger architectures, the 10M sequence dataset we use is not sufficiently large to use for pre-training from random initialization.

6 Discussion and future work

Using a synthetic dataset, we demonstrate that pre-caching can indeed be learned by a transformer model. On the other hand, our experiments with natural language suggest that the breadcrumbs hypothesis is more explanatory for that setting, especially with smaller models, but that the importance of pre-caching increases with scale.

If the myopia gap is indeed not large in practice, there may be several applications of myopic training. We hypothesize that myopic transformers may have advantages in terms of safety and/or interpretability—it may be easier to understand what a model is doing if we know that everything that it is computing on each forward pass is directly towards the goal of predicting the immediate next token. For example, as seen in Appendix D.3, myopic models may not be able to make use of computation on the forward passes of filler tokens in the sense of Pfau et al. (2024).

Another possibility is that of automatically swapping in a locally myopic model (Section 5.1) on forward passes where we detect it is beneficial to sacrifice future performance in favor of immediate next-token accuracy (for example, on especially important tokens, or near the end of a text). We leave these possible applications to future work.

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